rfsurrogates: how a contribution flows

A surrogate model needs a few hundred full-wave solves. One desktop does 50 solves in about 5 hours. The models and the data are open.

What the project needs

One full-wave solve: about 11 min on a desktop. One surrogate query: 4.5 ms.

New device families	A parameter box and a textbook model of the response.
Other solvers	HFSS, CST or openEMS, by export and import.
Independent checks	Another person tests the records and the method.

Open models, open data

An open alternative to closed commercial models.

	rfsurrogates	Closed models
Code, weights	open, Apache-2.0	closed
Data	open, CC-BY-4.0	not published
Who adds data	contributors	the owner

Family 1: 261 Palace solves. Held-out passband |S21| error:
0.52 dB for the trained model, 2.86 dB for the textbook prior alone.

1 Contributor solves designs

on their own machine, to their own budget

```
python tools/contribute.py go \
    --family edge_coupled_bpf --n 20
```

First run Draws designs from the parameter box.
Opens a claim issue on GitHub.
Claim gate: the maintainer accepts

Second run Builds each case.
Geometry gate: rejects a wrong layout

Solves each case on this machine.
Checks each record.
Opens one pull request.

Solver Palace: runs automatically.
HFSS, CST or openEMS: the tool exports the geometry. You solve it. The tool reads the Touchstone file back.

2 CI verifier

on the pull request

GitHub Actions runs the checker again on every record.

PASS

NEEDS REVIEW

FAIL

3 Maintainer merges

records are pooled

The records go to the Hugging Face dataset

`rfsurrogates/samples`

and to the data index.
Each record names its contributor.

4 Retrain, publish

open weights

Retrain when a family has about 25 % more records.

New weights go to Hugging Face.

Anyone can use them:

- browser demo
- inverse design
- download

more contributors → **more records** → **better models** → **more device families**